Experiment 2: Average Price of Products Using NumPy

Aim

To calculate the average price of all products sold during the past month using NumPy arrays.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a 3×3 NumPy array representing sales prices for three products across three sales records.
4. Calculate the average of all array elements using np.mean().
5. Display the average price.
6. Stop the program.

Code

```
import numpy as np

# Sales prices of 3 products across 3 sales records

sales_data = np.array([

   [100, 150, 200],

   [250, 300, 350],

   [120, 180, 240]

])

# Calculate the average price

average_price = np.mean(sales_data)

print("Average price of all products:",

    average_price)
```

Input

100 150 200

250 300 350

120 180 240

Output

Average price of all products: 210.0

```
========= RESTART: C:\Users\sivak\Desktop\FODS LAB experiments\exp2.py =========
Average price of all products: 210.0
```

Result

Thus, the average price of all products was calculated using the NumPy mean() function, and the average price was found to be 210.00.